



# S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

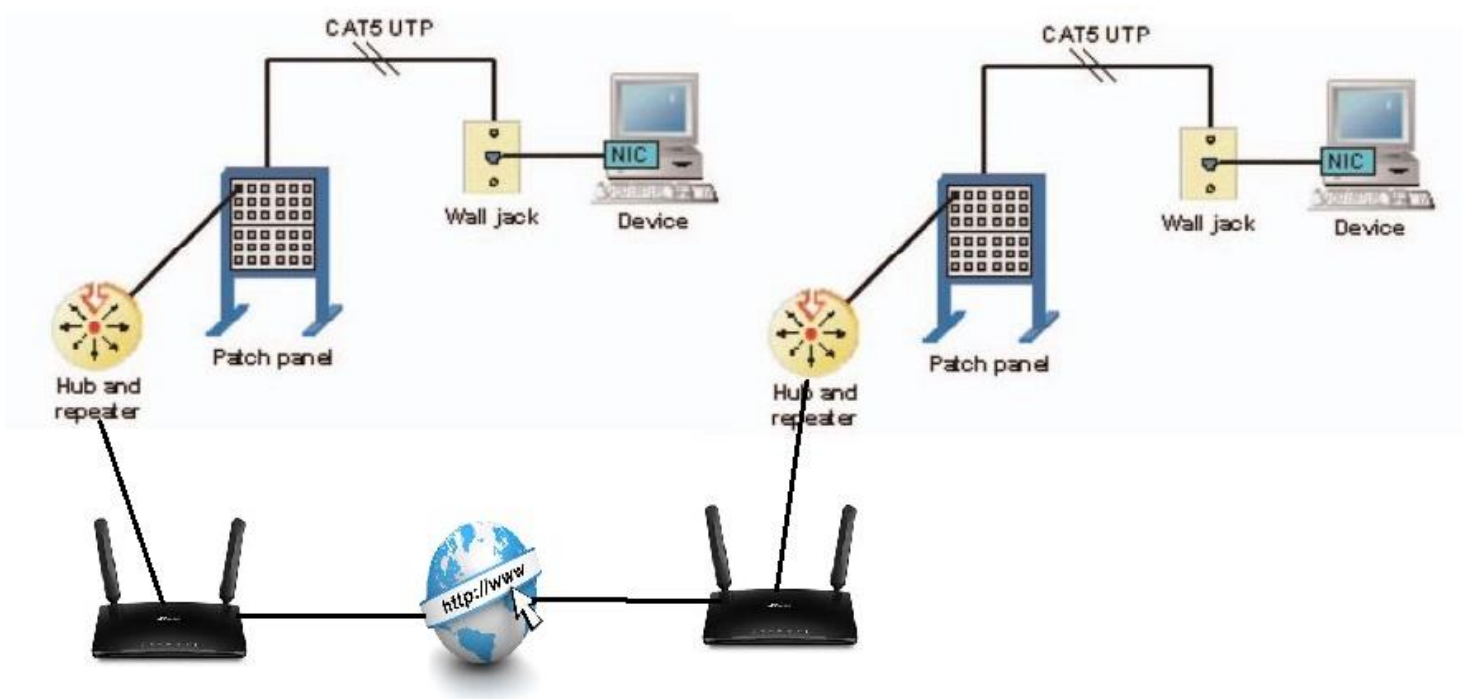
Part -2 (Paper-4)

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## INTERNET

### ❖ WHAT IS INTERNET?

- The Internet is a global network of interconnected computer networks that communicate with each other using standard communication protocols such as TCP/IP.
- In simple words, a network of networks connected worldwide is called the Internet. OR  
Collection of networks / interconnected of networks is called internet.
- The Internet allows two different networks computers and other devices to exchange information and communicate with each other.
- It provides many services such as:
  - ✓ E-mail
  - ✓ Chat and instant messaging
  - ✓ Video and audio conferencing
  - ✓ Online education
  - ✓ Online banking
  - ✓ Money transfer
  - ✓ Online shopping
  - ✓ Food ordering
  - ✓ Ticket booking
  - ✓ Entertainment
  - ✓ Online gaming, etc.



## ➤ Advantages of Internet

- **E-mail** - With e-mail you can send and receive instant electronic messages, which works like writing letters. Your messages are delivered instantly to people anywhere in the world.
- **24 hours a day** - 7 days a week - Internet is available, 24x7 days for usage
- **Information** - Biggest advantage internet is offering of users just about every subject, Encyclopedia, government law and services, trade fairs and conferences, market information, new ideas and technical support using search engines like google, yahoo, MS bing, DuckDuckGO,.... etc.
- **Online Chat** - You can access many 'chat rooms' on the web that can be used to meet new people, make new friends, as well as to stay in touch with old friends.
- **Services** - Many services are provided on the internet like net banking, job searching, purchasing tickets, hotel reservations, guidance services, Food order,...etc
- **Communities** - Communities of all types (different country) communicate on the internet.
- **E-commerce** – By Internet you can also shop online. E.g. – amazon, flipkart,.....

- **Entertainment** - : Internet provides facility to access wide range of Audio/Video songs, plays films. Many of which can be downloaded. One such popular website/app is YouTube.
- **Software Downloads** - You can freely download innumerable softwares like utilities, games, music, videos, movies, etc from the Internet.

## ➤ **Disadvantage/Limitations of Internet**

- **Theft of Personal information** - Electronic messages sent over the Internet can be easily snooped and tracked, revealing (खुलासा होना) who is talking to whom and what they are talking about.
- **Negative effects on family communication** - Decrease in communication and feeling of togetherness among the family members.
- **Internet addiction** - There is some controversy over whether it is possible to actually be addicted to the Internet. E.g. – Game, reels
- **Children using the Internet** - Most parents do not realize the dangers involved when their children log onto the Internet. When children talk to others online, they do not realize they could actually be talking to a harmful person. E.g. Bluewheel game
- **Virus threat** - Computers are mainly getting these viruses from the Internet. Virus is a program which disrupts the normal functioning of your computer systems.
- **Spamming** (धोखा-धड़ी) - : It is often viewed as the act of sending unsolicited email. Most spam is commercial advertising, often for doubtful products, get-rich-quick schemes, or quasi-legal services.

## ❖ **History of the Internet/TCP/IP model**

### **1. ARPANET (1969)**

The history of the Internet began with **ARPANET**, developed by the U.S. Department of Defense's Advanced Research Projects Agency (ARPA).

- ARPANET was established in 1969.
- Its purpose was to connect computers at research institutions and universities.
- It is considered the foundation of the modern Internet.

## 2. First Message (1969)

On October 29, 1969, the first message was sent over ARPANET between UCLA and the Stanford Research Institute.

The intended message was “**LOGIN**”, but the system crashed after the first two letters, “LO”.

## 3. Development of TCP/IP (1970s)

In the 1970s, researchers developed the **TCP/IP** protocols.

- TCP/IP allows different computer networks to communicate with each other.
- It became the standard communication protocol for the Internet.

## 4. Internet Protocol Standardization (1983)

**On January 1, 1983, ARPANET officially adopted TCP/IP.**

This date is commonly considered the birth of the modern Internet because it allowed different networks to communicate using a common protocol.

## 5. Domain Name System (1984)

The **Domain Name System** (DNS) was introduced in 1984.

It made it easier to access websites using names such as example.com instead of numerical IP addresses.

## 6. World Wide Web (1989–1991)

The **World Wide Web** was invented by Tim Berners-Lee in 1989.

- The first website became available in 1991.
- The Web made it easier to access information through websites and hyperlinks.
- The Internet and the World Wide Web are not the same thing. The Web is one service that operates on the Internet.

## 7. Commercial Internet (1990s)

During the 1990s, Internet access expanded beyond research institutions.

- Internet Service Providers (ISPs) began offering Internet access to the public.
- E-mail, websites and online businesses became increasingly popular.
- The Internet became an important tool for education, communication and commerce.

## 8. Modern Internet (2000s–Present)

The Internet has continued to grow through broadband, Wi-Fi, smartphones, cloud computing and social media.

Today, it supports online education, digital payments, video conferencing, online shopping, entertainment and many other services.

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## Important Dates for Exams

### Year    Event

**1969**    ARPANET established

**1969**    First ARPANET message

**1983**    TCP/IP adopted by ARPANET

**1984**    DNS introduced

**1989**    World Wide Web invented

**1991**    First website became available

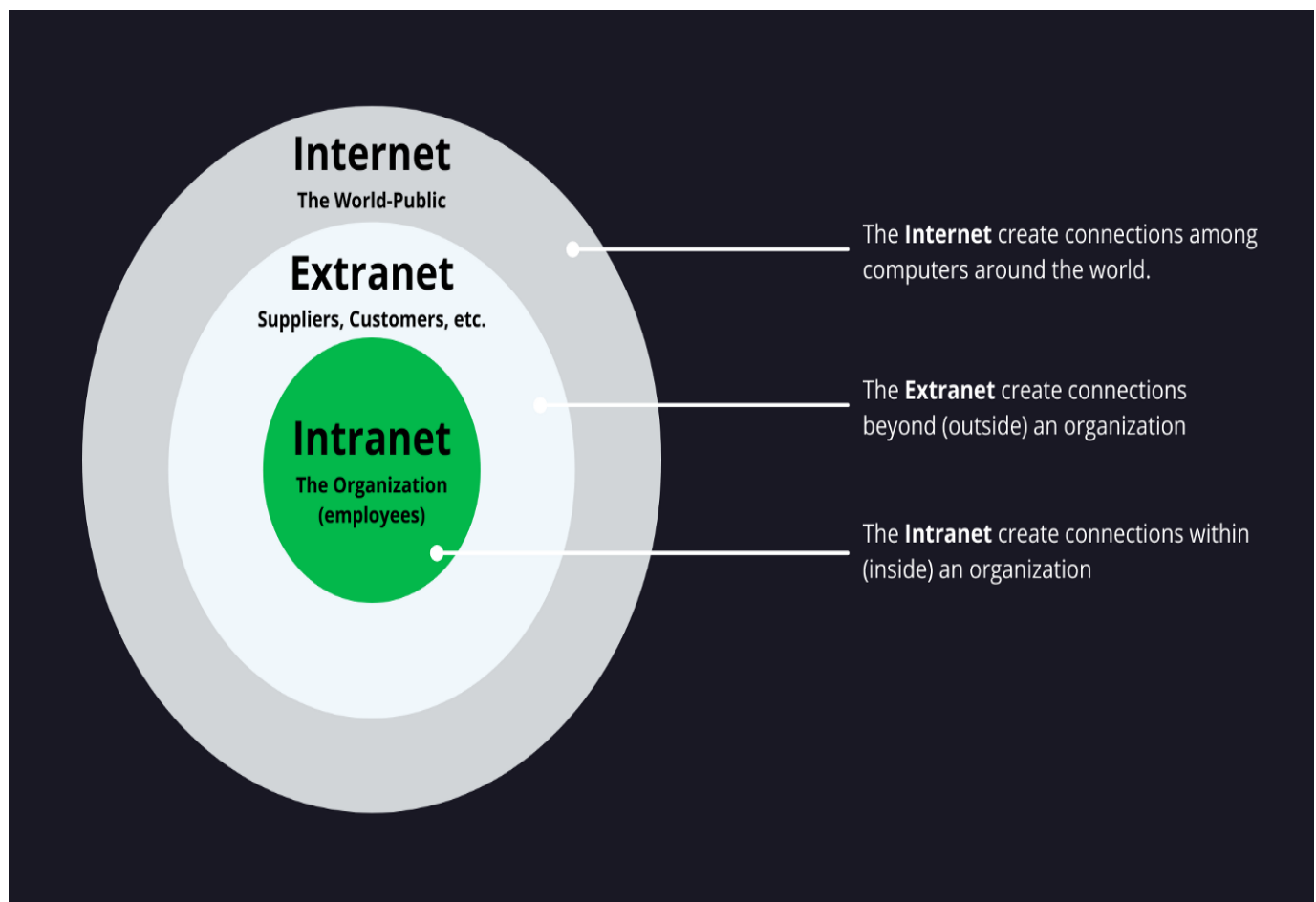
**1990s**    Internet became commercially available

## Q. What is Intranet ?

- An intranet is a private network used within an organization to securely share information, resources and services among authorized users.
- **Example:**  
A college may have an intranet through which teachers and staff access internal notices, student information, attendance systems and other internal resources.

## Q. What is Extranet ?

- An extranet is a private network that provides controlled access to authorized external users, such as customers, suppliers, business partners or other branches of an organization.
- **Example:** A company may allow its suppliers to access a particular part of its internal system to check orders or inventory.



## ❖ Difference B/w Internet, Intranet & Extranet

| Feature          | Internet                      | Intranet                              | Extranet                                                    |
|------------------|-------------------------------|---------------------------------------|-------------------------------------------------------------|
| <b>Nature</b>    | Public/global network         | Private network                       | Private network with controlled external access             |
| <b>Users</b>     | General public                | Authorized members of an organization | Authorized internal and external users                      |
| <b>Access</b>    | Generally publicly accessible | Restricted                            | Restricted and controlled                                   |
| <b>Coverage</b>  | Worldwide                     | Usually within an organization        | May connect organizations/branches over different locations |
| <b>Ownership</b> | No single owner               | Controlled by an organization         | Controlled by organization(s)                               |
| <b>Security</b>  | Security                      | Generally more                        | Generally more                                              |

|                       |                                                         |                                             |                                            |
|-----------------------|---------------------------------------------------------|---------------------------------------------|--------------------------------------------|
|                       | depends on services and configuration                   | restricted                                  | restricted and controlled                  |
| <b>Purpose</b>        | Public communication and information sharing            | Internal communication and resource sharing | Secure collaboration with external parties |
| <b>Example</b>        | Websites, e-mail, Google, YouTube                       | College/Company internal portal             | Supplier/customer business portal          |
| <b>Network Type</b>   | Can use LAN, MAN, WAN and other interconnected networks | Usually uses LAN/WAN technologies           | Can use LAN/WAN/VPN and other technologies |
| <b>Data Sharing</b>   | Public/general information                              | Within an organization                      | With authorized external parties           |
| <b>Authentication</b> | May or may not require login                            | Usually requires authentication             | Requires controlled authentication         |

## ❖ TCP/IP Model

- TCP/IP is reference (guide) model/ standard conceptual model.
- TCP/IP (Transmission Control Protocol/Internet Protocol) model is a networking model that explains how data is communicated between devices over a network or the Internet.
- The TCP/IP network model takes its name from two of its protocols, the Transmission Control Protocol (TCP) and the Internet Protocol (IP).
- TCP – Transfer the data over the internet. It divides the data into small parts and sends it to the destination through the network.
- IP – IP does Carry the address of data and reach the destination.
- It defines how data is prepared, transmitted, addressed, routed, and



delivered from one device to another.

- Old name of TCP/IP model is NCP model (Network control protocol).
- It is also referred to as the TCP/IP protocol suite and the Internet protocol suite/group.

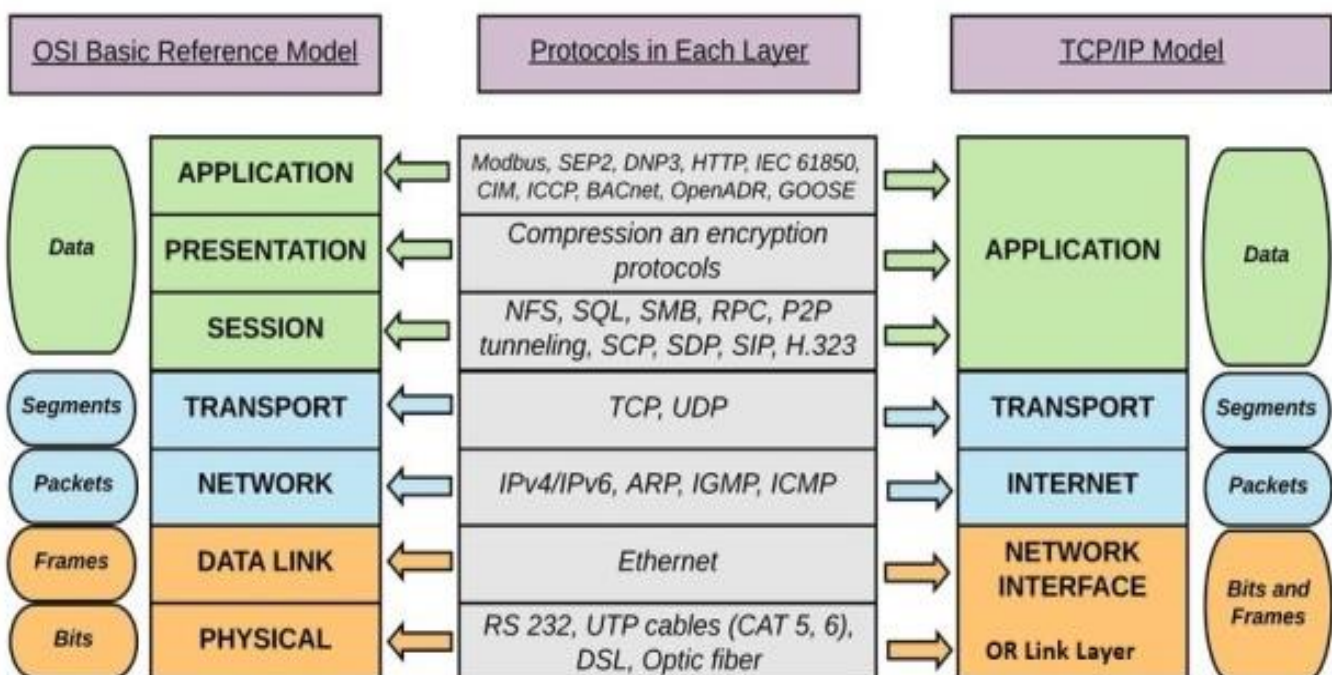
## Type of TCP/IP model

In original 1970's TCP/IP model had a total of 4 layer.

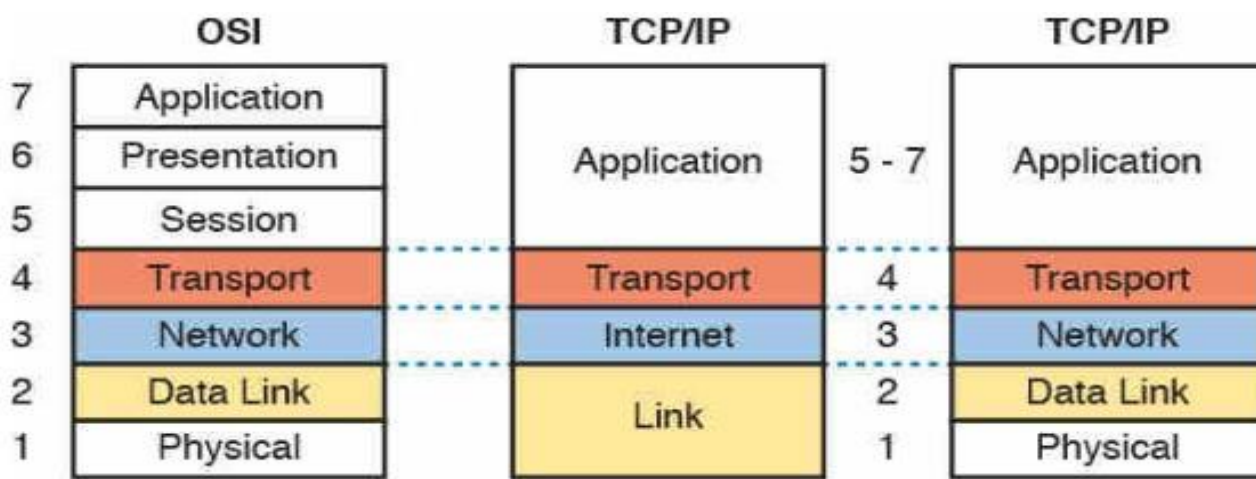
1. Application layer (Layer 4)
2. Transport layer (Layer 3)
3. Internet layer (layer 2)
4. Network interface layer / Link layer / Network Access Layer (layer 1)

The modern TCP/IP model is commonly represented as a five-layer model. In the 5-layer TCP/IP model, the Network Interface/Link layer of the 4-layer model is divided into two separate layers:

1. Application layer (Layer 5)
2. Transport layer (Layer 4)
3. Internet/Network layer (Layer 3)
4. Data Link layer (Layer 2)
5. Physical Layer (Layer 1)







## 4. Application Layer

- The Application Layer directly provides network services to user applications. It's where **network applications** can access network services. It contains protocols used by network applications for communication.
- **Network Application**
  - ✓ Web browser – chrome, firefox, edge,...
  - ✓ Email – Gmail, outlook
  - ✓ Chat/messaging – whatsapp, telegram
  - ✓ File transfer – FTP applications
  - ✓ Video Conferencing – Zoom, google meet
  - ✓ VoIP – Internet based calling applications
  - ✓ Remote access – Remote desktop, ssh, teamviewer, anydesk
- **Important protocols:**
  - ✓ **HTTP (HyperText Transfer Protocol):** Used to access and transfer web pages and web resources.
  - ✓ **SMTP (Simple Mail Transfer Protocol):** Used to send and transfer email from one mail server to another.
  - ✓ **POP3 (Post Office Protocol version 3):** Used by an email client to download emails from a mail server.
  - ✓ **IMAP (Internet Message Access Protocol):** Allows users to access and synchronize emails while keeping them on the mail server.
  - ✓ **FTP (File Transfer Protocol):** Used to upload and download files between computers over a network.
  - ✓ **DNS (Domain Name System):** Converts domain names such as www.google.com into IP addresses.
  - ✓ **Telnet:** A protocol used for remote command-line access to

another computer. It is **not secure**.

- ✓ **SSH (Secure Shell)**: A secure protocol used for remote login and executing commands on another computer.

**Note:** Telnet connects to a **remote computer**, not a local computer.

### 3. Transport layer

- Transport layers ensure that data sent by an application over the network arrives safely and correctly at the destination.
- The transport layer works around two main protocols:
  - ✓ **TCP (Transmission Control Protocol)** provides reliable, ordered, and error-checked data transfer between applications.
  - ✓ **UDP (User Datagram Protocol)** manages a simpler and faster data exchange without establishing a connectionless—mostly used in streaming. It is an unreliable protocol.

### 3. Internet layer

- The Internet Layer is responsible for logical addressing and routing packets between different networks.
- The Internet layer sends packets across multiple networks to reach their final destinations.
- These protocols work together to traverse data through the complex web of networks.
- The internet layer is directly linked with internet connections and operates through three main protocols:
  - ✓ **IP (Internet Protocol)** is the primary protocol to define IP addresses to route data packets across the Internet.
  - ✓ **ICMP (Internet Control Message Protocol)** reports errors. For example, it informs the network when a service is unavailable or a router cannot be reached. Example : ping command.

### 4. Network interface / Network access/ Link layer

- This layer is responsible for transferring data over the local network and deals with network hardware and transmission technologies.
- The network interface layer physically transfers data packets over the network.

- This layer is used in your computer's network card, cables, and switches that connect your devices.
- This layer not define any protocol.

### Examples:

- **Ethernet:** A common wired LAN technology.
- **Wi-Fi:** A wireless LAN technology.
- **PPP (Point-to-Point Protocol):** Used for communication between two network points.
- **Token Ring:** An older LAN technology that uses token passing.

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